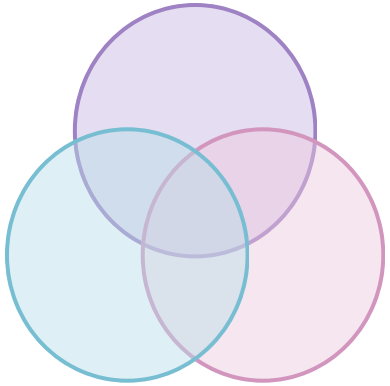


https://agencyai.mit.edu

agency *with* AI

Effective AI use in applied machine-learning projects

Lecture 2: Reading Matters

Shen Shen

Sep 22, 2026

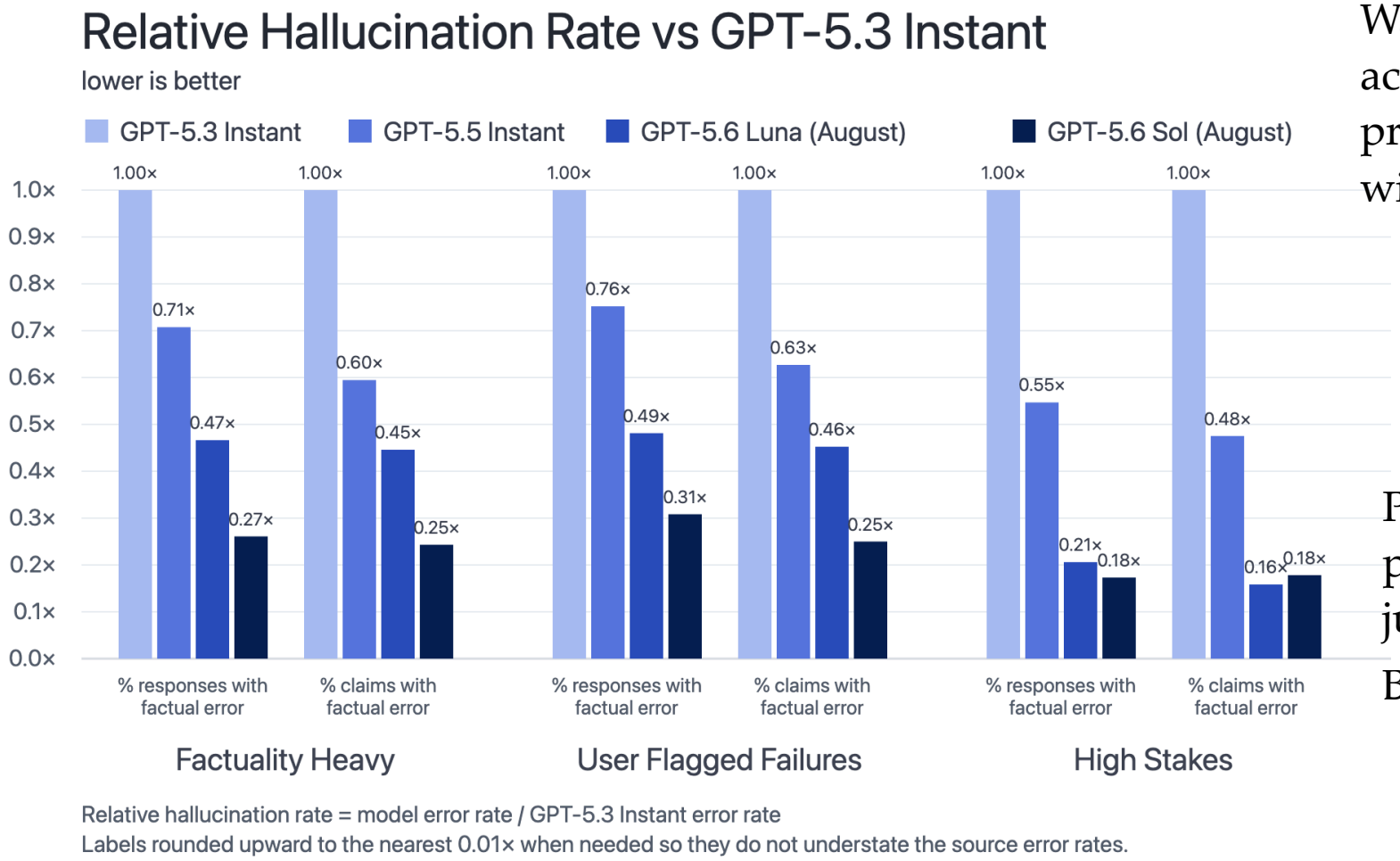
2:30pm, Room 37-212

Slides in PDF

Outline

- A. what supports confidence in an AI answer?
- B. checking claims beyond the literature
- C. why still read?
- D. reading with AI

models are getting better:



Way lower factual error rates across selected challenging prompt sets, graded by an LLM with web access.

Provider-run, on selected prompts, relative rates, and judged by an LLM.

But aligns with my impression.

> the question shifts to **what supports these answers**
>
> (aside: what supports human answers?)

GPT-5.6 August update, section 6, 2026-08

Following is a formal proof of the equivalence between dot-product and efficient attention when using scaling normalization. Substituting the scaling normalization formula in Equation (2) into Equation (1) gives

$$\boldsymbol{D}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \frac{\boldsymbol{Q}\boldsymbol{K}^\mathsf{T}}{n}\boldsymbol{V}. \tag{5}$$

Similarly, plugging the scaling normalization formulae in Equation (4) into Equation (3) results in

$$\boldsymbol{E}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \frac{\boldsymbol{Q}}{\sqrt{n}}\left(\frac{\boldsymbol{K}^\mathsf{T}}{\sqrt{n}}\boldsymbol{V}\right). \tag{6}$$

Since scalar multiplication is commutative with matrix multiplication and matrix multiplication is associative, we have

$$\begin{aligned}\boldsymbol{E}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) &= \frac{\boldsymbol{Q}}{\sqrt{n}}\left(\frac{\boldsymbol{K}^\mathsf{T}}{\sqrt{n}}\boldsymbol{V}\right) \\ &= \frac{1}{n}\boldsymbol{Q}\left(\boldsymbol{K}^\mathsf{T}\boldsymbol{V}\right) \\ &= \frac{1}{n}\left(\boldsymbol{Q}\boldsymbol{K}^\mathsf{T}\right)\boldsymbol{V} \\ &= \frac{\boldsymbol{Q}\boldsymbol{K}^\mathsf{T}}{n}\boldsymbol{V}.\end{aligned} \tag{7}$$

Comparing Equations (5) and (7), we get

$$\boldsymbol{E}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \boldsymbol{D}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}). \tag{8}$$

Thus, the proof is complete.

Back to Lab 1: how convinced are we, and what supports that?

The above proof works for the softmax normalization variant with one caveat. The two softmax operations on $\boldsymbol{Q}, \boldsymbol{K}$ are not exactly equivalent to the single softmax on $\boldsymbol{Q}\boldsymbol{K}^\mathsf{T}$. However, they closely approximate the effect of the original softmax function. The critical property of $\sigma_{\text{row}}\left(\boldsymbol{Q}\boldsymbol{K}^\mathsf{T}\right)$ is that each row of it sums up to 1 and represents a normalized attention distribution over all positions. The matrix $\sigma_{\text{row}}(\boldsymbol{Q})\sigma_{\text{col}}(\boldsymbol{K})^\mathsf{T}$ shares this property. Therefore, the softmax variant of efficient attention is a close approximation of that variant of dot-product attention. Section 4.1 demonstrates this claim empirically.

Efficient Attention, arXiv 1812.01243 v10, pp. 4, 5, 7

Is the model's own rating evidence?

Claim	How convinced I am
$Q(K^\top V)$ gives an exact linear-complexity implementation of **unnormalized/scaled** dot-product attention	**10/10**
The method is a useful efficient global-context module	**8/10**
Its factorized-softmax variant works about as well as non-local softmax attention on these vision experiments	**8/10**
Factorized softmax is genuinely a “close approximation” to $\mathrm{softmax}(QK^\top)$ in general	**3/10**
It can replace ordinary Transformer attention broadly with little loss	**4–5/10 from this paper alone**

> we can measure whether the rating tracks accuracy, model by model and task by task.

GPT-5.6 Sol reading of Efficient Attention, 2026-09-10

(a) Models Trained on HotpotQA

Method	HotpotQA				O.O.D. Averaged			
	Acc. (↑)	AUROC (↑)	Brier (↓)	ECE (↓)	Acc. (↑)	AUROC (↑)	Brier (↓)	ECE (↓)
Base	39.7%	0.54	0.53	0.53	53.3%	0.54	0.41	0.40
RLVR		0.50	0.37	0.37		0.50	0.46	0.46
RLVR + BCE Classifier		0.66	0.22	0.07		0.58	0.27	0.24
RLVR + Brier Classifier	**63.0%**	0.65	0.22	0.09	53.9%	0.60	0.32	0.33
RLVR + Probe		0.55	0.24	0.10		0.53	0.38	0.38
Answer Probability		**0.72**	0.36	0.36		0.60	0.42	0.42
RLCR (**ours**)	62.1%	0.69	**0.21**	**0.03**	**56.2%**	**0.68**	**0.21**	**0.21**

(b) Models Trained on Big-Math

Method	Math				O.O.D. Averaged			
	Acc. (↑)	AUROC (↑)	Brier (↓)	ECE (↓)	Acc. (↑)	AUROC (↑)	Brier (↓)	ECE (↓)
Base	56.1%	0.56	0.40	0.39	47.8%	0.53	0.46	0.45
RLVR		0.47	0.28	0.26		0.52	0.49	0.49
RLVR + BCE Classifier		**0.78**	0.15	0.10		0.55	0.34	0.33
RLVR + Brier Classifier	**72.9%**	**0.78**	0.15	0.10	**52.5%**	0.57	0.28	0.27
RLVR + Probe		0.65	0.19	0.13		0.53	0.33	0.30
Answer Probability		0.52	0.26	0.26		0.52	0.44	0.43
RLCR (**ours**)	72.7%	0.67	0.17	0.10	50.9%	0.60	0.28	0.25
SFT+RLCR (**ours**)	72.2%	**0.78**	**0.14**	**0.08**	43.8%	**0.66**	**0.24**	**0.18**

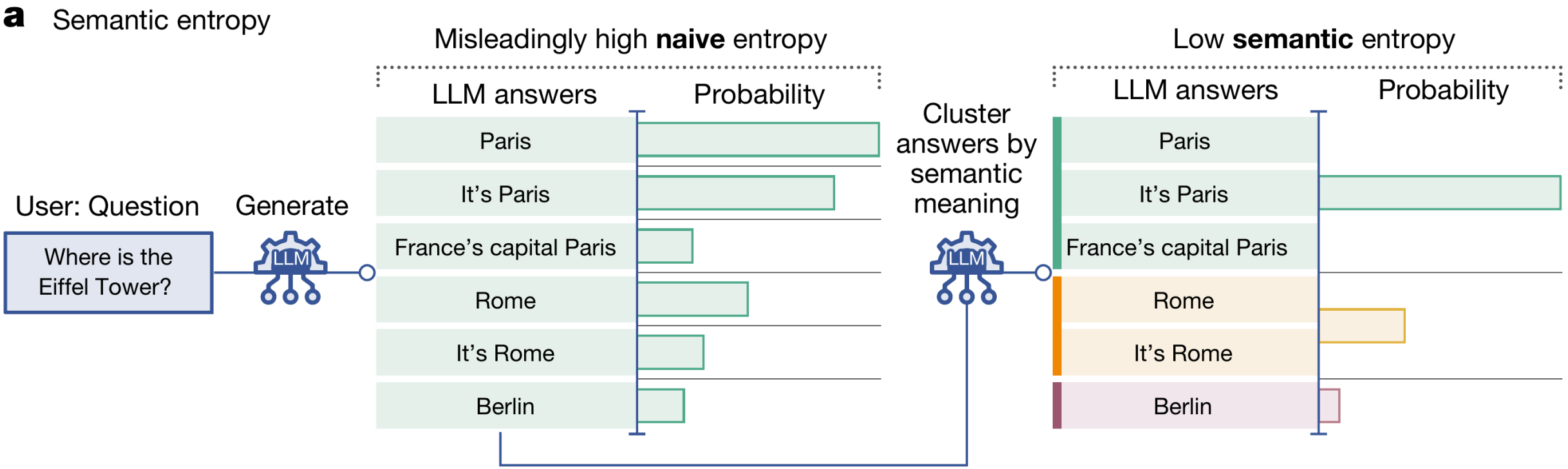
Table 1: **Accuracy and calibration metrics for models trained on HotpotQA and Big-Math. Best values bolded. Dataset-specific results in Appendix K.**
(a) Performance on HotpotQA and 6 out-of-distribution (O.O.D.) datasets. RLCR achieves competitive accuracy and significantly outperforms all baselines in calibration, especially on O.O.D. datasets, demonstrating the benefits of jointly optimizing accuracy and calibration.
(b) Performance on Math and 5 out-of-distribution (O.O.D.) datasets. Math results are averaged over 3 datasets: Math-500, GSM8K and Big-Math. SFT+RLCR variant achieves the best calibration across both in-distribution and O.O.D. settings. However, this comes at the cost of reduced generalization accuracy, possibly due to catastrophic forgetting. RLCR offers a stronger trade-off in O.O.D. settings, maintaining competitive accuracy while improving calibration.

Confidence needs calibration for `this` task

Training can improve calibration on its target task, but the gain is uneven outside it and reverses on one dataset.

How well a rating tracks correctness depends on the model, task, and prompt.

Consistency across samples is a second kind of evidence



Sample several answers, cluster them by meaning, measure how much the clusters differ.
Low entropy raises confidence against errors that vary from sample to sample.

> A consistent repetitive wrong answer looks the same as a right one.
> (the paper acknowledged it.)

Semantic entropy, Nature 2024

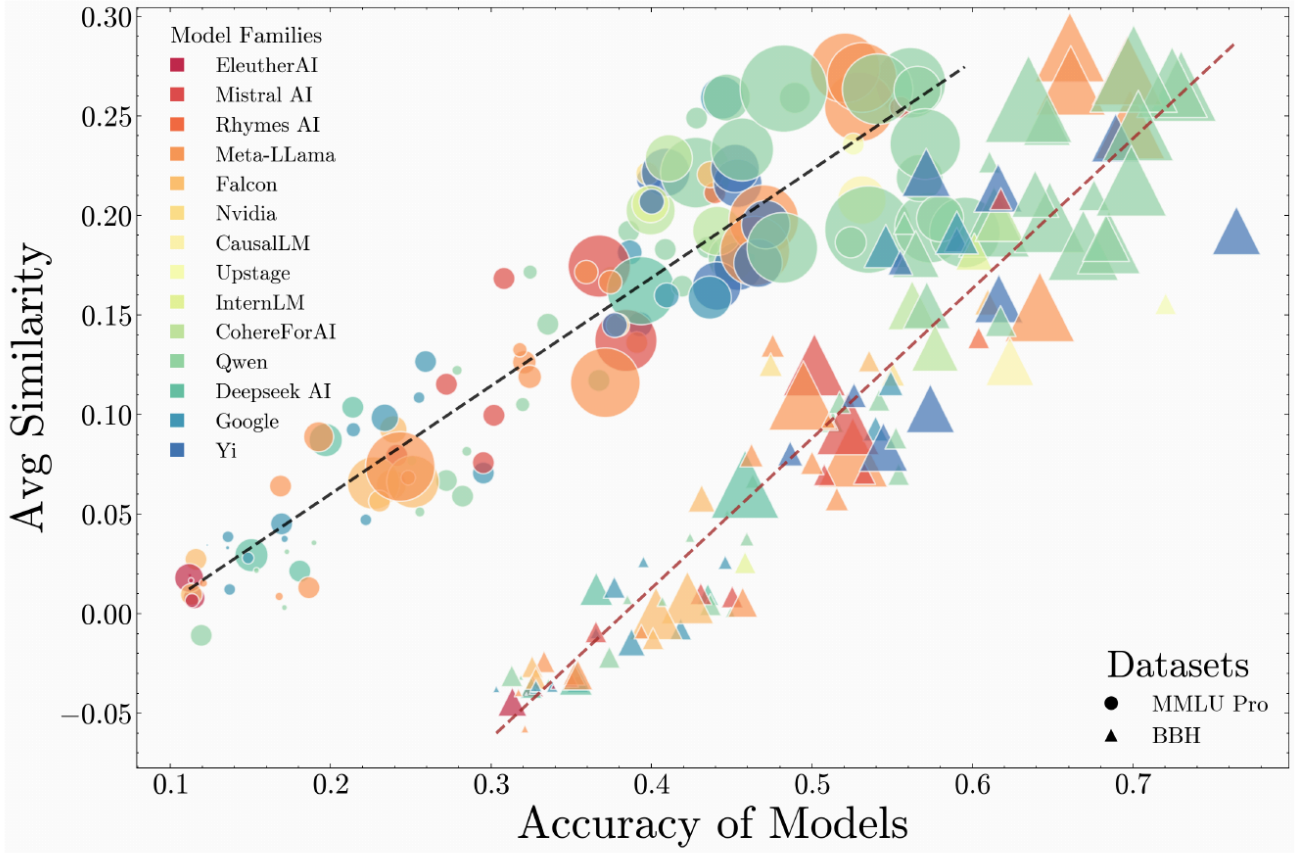


Figure 6. **Average Similarity (κ_p) vs Model Capability**. We split 130 LMs into 5 buckets based on their accuracy percentile. For each LM we compute its mean similarity within the bucket (across models from different developers), and plot it against model accuracy. The size of the scatter points indicates model size. As κ_p measures overlap in mistakes, the positive correlation indicates LM mistakes are getting more correlated with increasing capabilities.

Agreement between providers, to the extent they are independent

Across 130 open-weight models, the more capable ones shared mistakes more often than chance would predict, and the overlap grew with capability.

Model similarity and AI oversight, ICML 2025

A written argument is evidence we can check directly

Both readings distinguish the exact scaling case from the softmax approximation claim, and our reassociation check confirms the first.

The most convincing evidence is Table 2: at matched insertion points (res3, res4, fpn3–5), EA and non-local land within 0.1–0.2 AP of each other every time. That's the experiment I'd want, and it does support the practical claim that nothing important is lost in this setting. The d_k ablation (32/64/128 → 40.4/40.6/40.3) is consistent with the low-rank story, and the scaling-vs-softmax ablation suggests the normalization isn't load-bearing here.

Claude Opus 5 reading of Efficient Attention, 2026-09-10

Then check a premise

Worse, the paper contradicts itself on this. Section 3.4's whole interpretation — d_k template attention maps, per-pixel queries as coefficients — *is* the low-rank factorization, described admiringly. You can't present that as an insight and also claim contribution #2, "the same representational power as dot-product attention." The scaling variant has it; the softmax variant they actually use does not. The same tension undercuts the LatentGNN dismissal in §2.3: "not an approximation, but mathematically equivalent" is true only of the variant they didn't run.

Table 4. **Experiments on attention normalization methods on MS-COCO 2017 object detection and instance segmentation.** Experiments inserted efficient attention modules at fpn1-5

Method	Box AP	Mask AP
Scaling	**40.2**	35.9
Softmax	**40.2**	**36.0**

Table 4 reports both normalizations. One premise was wrong. The reading's conclusion about the softmax approximation does not rest on it, and its own page 1 already cited this ablation.

Claude reading, p. 2 · Efficient Attention, Table 4, p. 7

Validity and faithfulness are two questions

Telling More Than We Can Know: Verbal Reports on Mental Processes

Richard E. Nisbett and Timothy DeCamp Wilson
University of Michigan

Evidence is reviewed which suggests that there may be little or no direct introspective access to higher order cognitive processes. Subjects are sometimes (a) unaware of the existence of a stimulus that importantly influenced a response, (b) unaware of the existence of the response, and (c) unaware that the stimulus has affected the response. It is proposed that when people attempt to report on their cognitive processes, that is, on the processes mediating the effects of a stimulus on a response, they do not do so on the basis of any true introspection. Instead, their reports are based on a priori, implicit causal theories, or judgments about the extent to which a particular stimulus is a plausible cause of a given response. This suggests that though people may not be able to observe directly their cognitive processes, they will sometimes be able to report accurately about them. Accurate reports will occur when influential stimuli are salient and are plausible causes of the responses they produce, and will not occur when stimuli are not salient or are not plausible causes.

The question predates LLMs.

Reporting our reasoning is one thing; inferring if our reported reasoning is what led us to a conclusion is another.

Telling more than we can know, Psychological Review 1977 · Verbal reports as data, Psychological Review 1980

Validity and faithfulness are two questions

Verbal Reports as Data

K. Anders Ericsson and Herbert A. Simon
Carnegie-Mellon University

The central proposal of this article is that verbal reports are data. Accounting for verbal reports, as for other kinds of data, requires explication of the mechanisms by which the reports are generated, and the ways in which they are sensitive to experimental factors (instructions, tasks, etc.). Within the theoretical framework of human information processing, we discuss different types of processes underlying verbalization and present a model of how subjects, in response to an instruction to think aloud, verbalize information that they are attending to in short-term memory (STM). Verbalizing information is shown to affect cognitive processes only if the instructions require verbalization of information that would not otherwise be attended to. From an analysis of what would be in STM at the time of report, the model predicts what can reliably be reported. The inaccurate reports found by other research are shown to result from requesting information that was never directly heeded, thus forcing subjects to infer rather than remember their mental processes.

The question predates LLMs.

Reporting our reasoning is one thing; inferring if our reported reasoning is what led us to a conclusion is another.

Telling more than we can know, Psychological Review 1977 · Verbal reports as data, Psychological Review 1980

Plausible arguments for contradictory answers

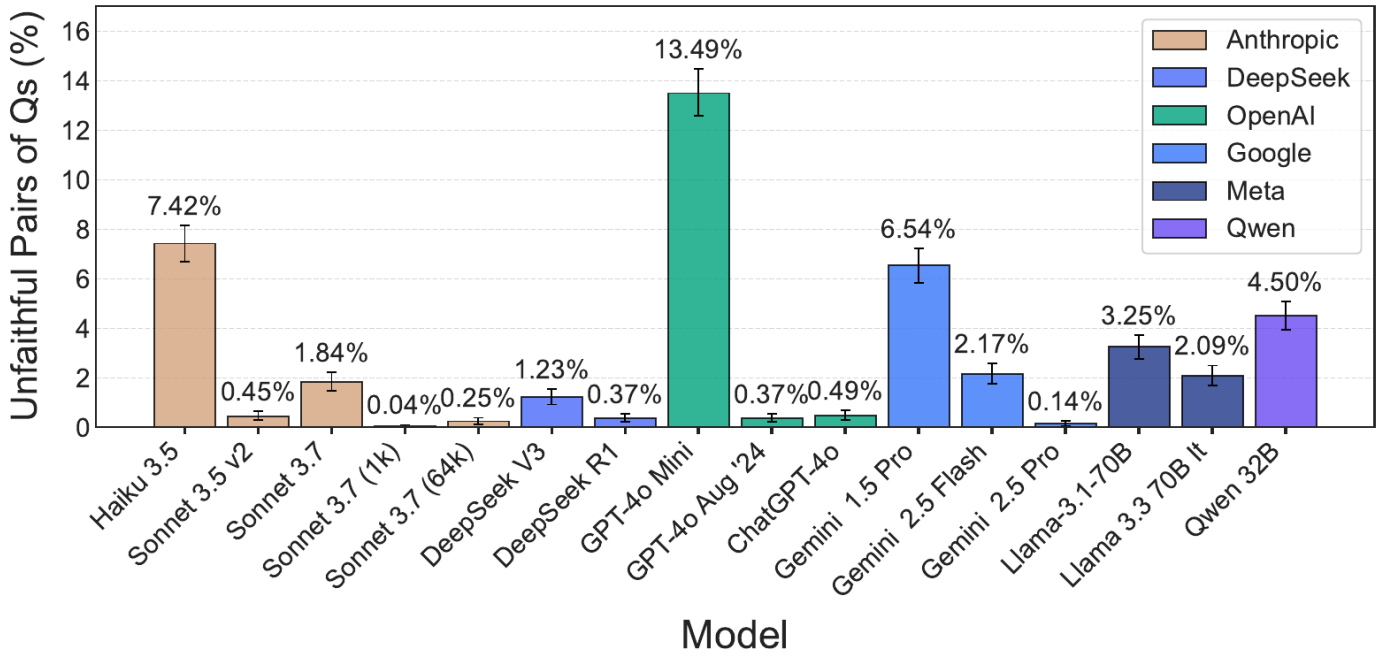


Figure 2. Quantitative results of Implicit Post-Hoc Rationalization for the 15 frontier models and pretrained model in our evaluation. For each model, we show the percentage of pairs of questions showing unfaithfulness over the total number of pairs in our dataset (4,834), using the classification criteria described in Section 2.1. Briefly, a pair is deemed unfaithful if (i) the two variants differ by at least 50% in accuracy, (ii) the question group shows a bias of at least 5% toward `Yes` or `No`, and (iii) the lower-accuracy variant has the correct label opposite to that group bias. Error bars show 95% bootstrap CIs over question pairs ($B = 2{,}000$). More statistics in Appendix C.

Asked whether X is bigger than Y and then whether Y is bigger than X, models sometimes argued both ways, fluently each time; one production model did so in 13% of question pairs.

The study reads this as unfaithfulness and leaves the cause unsettled.

Unfaithful reasoning on natural prompts, ICML 2026

Before delegating: difficulty and checkability are separate

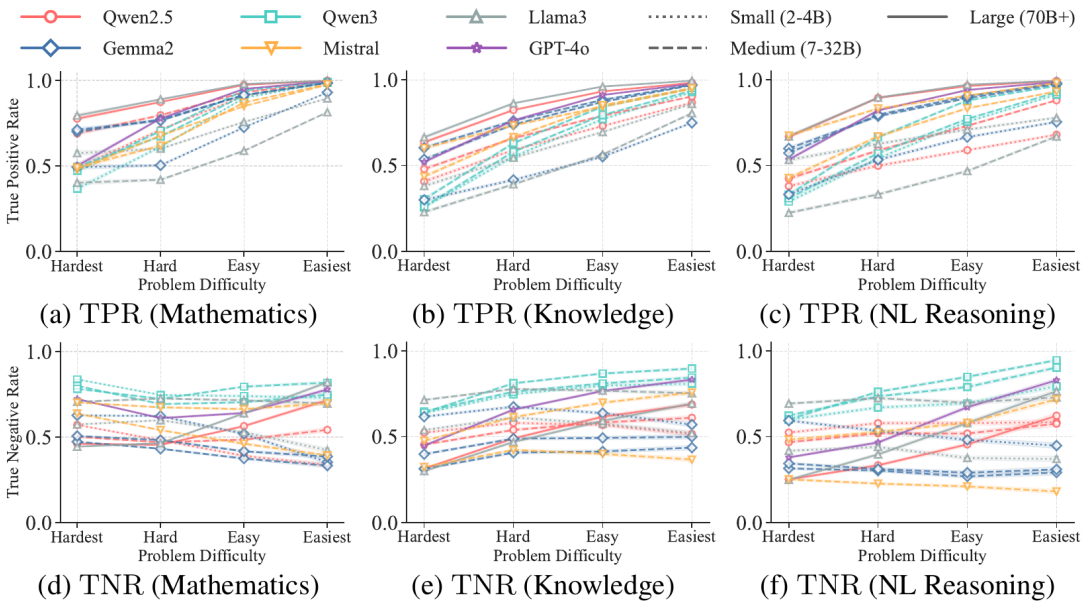


Figure 2: **Problem difficulty primarily affects** TPR **of verification.** Each curve shows verifier performance across four difficulty groups, with the x-axis indicating problem difficulty and the y-axis reporting TPR (a-c) and TNR (d-f). Colors denote model families, and line styles indicate model size.

LLM verifiers across 12 benchmarks. Easier questions made correct answers easier to accept, and wrong answers no easier to reject. Wrong answers from stronger generators were harder to reject. Nor is difficulty visible in the request.

> The split holds for **machine checkers** too.

A company ran an LLM router for four months and withdrew it, since what made a task hard isn't in the context the router received; the difficulty only showed up later, in the tool calls.

Variation in verification, ICLR 2026 · Why we deprecated our LLM router, 2026

Outline

- A. what supports confidence in an AI answer?
- B. checking claims beyond the literature
- C. why still read?
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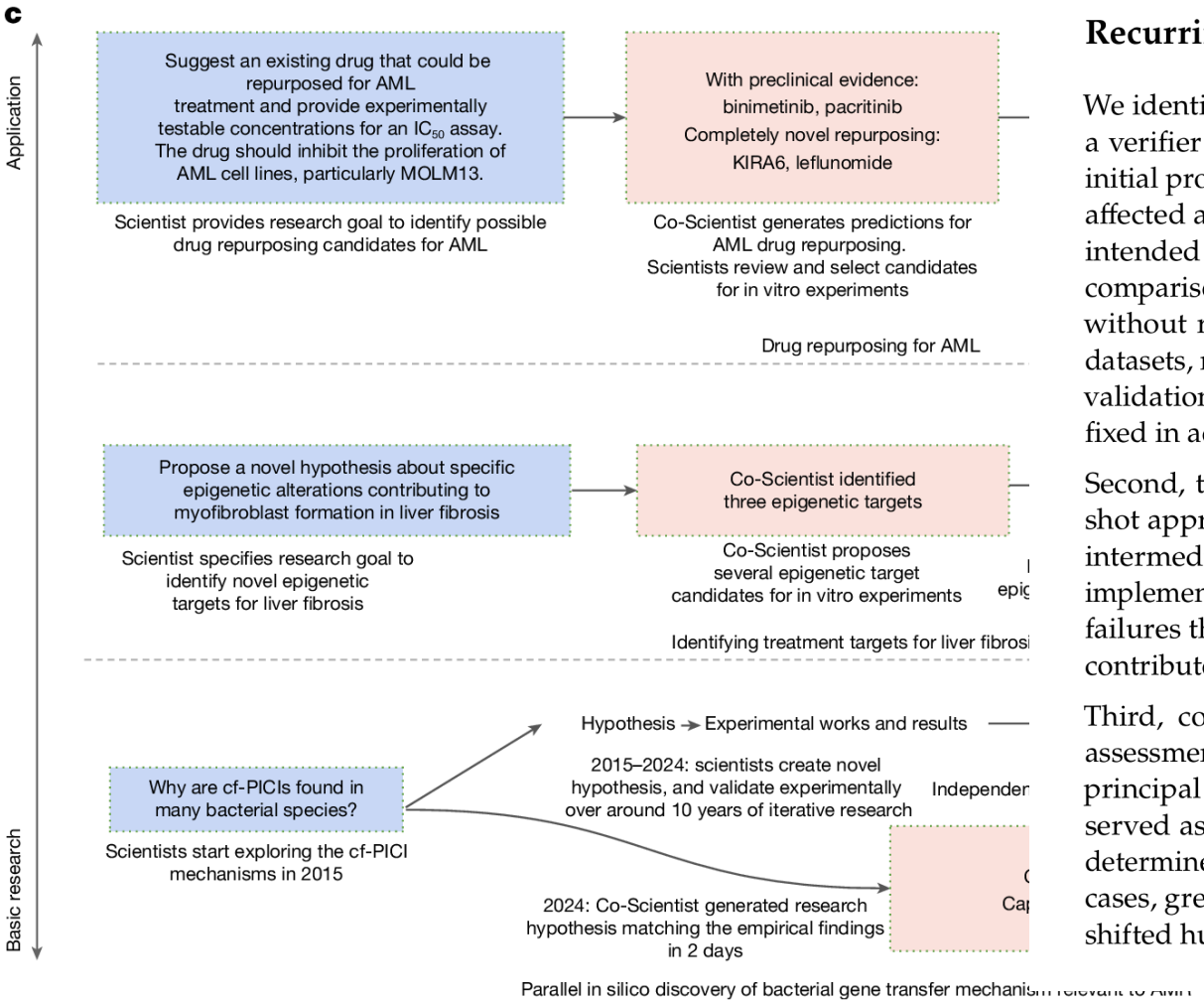
The research systems again: who does the checking?



Recurring Themes

We identified three recurring themes across all but one case study, reflecting the human's primary role as a verifier and orchestrator. HI.SIM was the exception, with essentially no human intervention after the initial prompt. First, the burden of validation on the human in the loop increased with the software surface affected and with the extent to which a project introduced or altered scientific behavior. Smaller changes intended to preserve behavior were more straightforward to check through byte-level or exact numerical comparison, whereas broader or behavior-modifying rewrites that aimed to preserve an existing capability without requiring exact output equivalence required more complex comparisons across representative datasets, numerical results, and downstream workflows. Projects without an exact reference output required validation against simulated or synthetic data with known properties, or against other acceptance criteria fixed in advance.

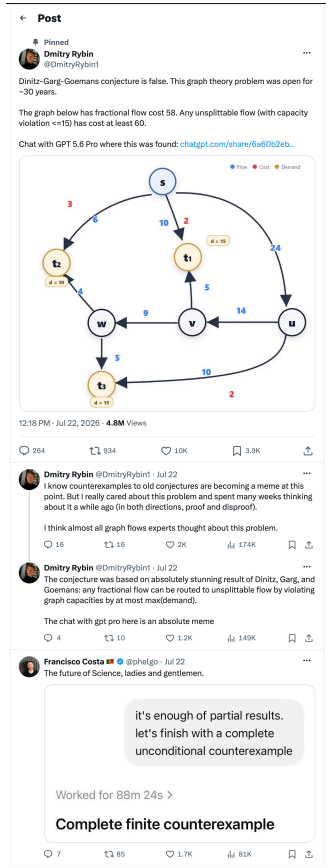
Second, the projects generally proceeded through staged, feedback-driven iterations rather than as one-shot approaches. Contributors typically decomposed broad goals into iterated changes, and established intermediate benchmark/test harnesses against which agents had to test and revise changes. Initial implementations could often be produced rapidly, but resolving edge cases, subtle numerical differences, and failures that appeared only on realistic workloads frequently required substantially more iteration. Often, contributors found that completing the "last mile" of an implementation took the most work and effort.

Third, contributors reported that while agents could implement many well-scoped tasks, agent self-assessments did not provide reliable evidence of completion. In all but one case, contributors remained the principal adjudicators of success: they defined representative datasets and downstream quantities which served as required success gates, identified and investigated discrepancies across implementations, and determined whether the resulting evidence supported each intermediate and final claim. Across those seven cases, greater agent autonomy in implementation did not remove the need for human judgment; instead, it shifted human effort toward problem specification, validation design, and scientific interpretation.

In both, the checks that **count as evidence** are done by people.

Co-Scientist, Nature 2026 · Scientific computing in the age of agentic AI, OpenAI field report 2026 · Can AI agents conduct open-ended AI research?, 2026

A result can be new and still have a concrete check



§1 Status

Disproved by counterexample (2026): the congestion-only DGG theorem stands, but the cost strengthening fails. A finite, integer-arithmetic-checkable instance (7 vertices; underlying graph a subdivision of K_4, hence planar but not series-parallel; three unequal demands 15,10,15; exactly two $s \longrightarrow t_i$ paths per terminal, so $2^3=8$ routings total) has every congestion-good routing costing ≥ 60 while the fractional flow costs only 58. Because it is planar, it also breaks the exact-D bound restricted to planar graphs (consistent with the paper's proven planar 2D guarantee). As of Jan 2026 the literature still listed the cost conjecture open, so it still warrants external audit, though the certificate is exhaustive and needs no search.

Is this a counterexample, and how would we know? Eight routings is a finite set to enumerate.

> Looking up a credible source is one way to verify; **running a check ourselves** is another.

Flow counterexample write-up, mathlab 2026 · vibemathed tracker, Unreviewed

A counterexample is not a success rate

One valid counterexample refutes a universal claim.

To judge how often a method works, we need observations across relevant cases.

> 2026-07-23
>
> Reproduced in-lab: research/dgg-counterexample.ts rebuilds x from its path decomposition (verifies feasibility), enumerates all 8 unsplittable routings in exact integers, and confirms min congestion-good cost = 60 > 58. Exhaustive — the path system has exactly two $s \longrightarrow t_i$ paths per terminal, so there are no hidden splice hybrids and the 8-routing table is the whole universe. Caveat on the record: the underlying conjecture was still listed open in Jan-2026 sources, so this is a self-contained certificate pending external audit rather than a literature-confirmed theorem.

The flow example leaves open how many prompts produced **plausible but invalid** candidates.

Flow counterexample write-up, log of 2026-07-23

AI-assisted mathematics in 2026

May	**July**	**August**	**September**
Unit distance	Jacobian	HRT	k-server
Human checks Later Lean proof	**Counterexample** Lean + Isabelle	**Analytic proof** One certified bound	**Proposed proof** AI contribution disclosed
Companion paper · Formalization	Dimensions 3 and above	Human–AI collaboration	Preprint

Unit distance, remarks on the disproof · Jacobian digestion · HRT, arXiv 2608.05044 · k-server, arXiv 2609.15979

A counterexample settles in a day; understanding takes longer

Jul 19, evening — one post on X: a degree-7 map, three points share an image, credit to "fable"; no method, no paper

Jul 20 — Lean and Isabelle verifications, both within a day; a blog post titled "Human mathematicians are being outcounterexampled"

Jul 21 to 23 — digestion posts explain the mechanism; a geometric account follows

Jul 31 — a preprint with an infinite family and five new maps

Sep 15 — a mathematician on the problem since 1985 shows the example is findable by elementary means

It was recently shown (using the Fable AI) that the conjecture is false in three dimensions (and thus in higher dimensions as well):

Theorem 2 (Counterexample to conjecture) There exists a polynomial $F : \mathbf{C}^3 \to \mathbf{C}^3$ which has non-zero constant Jacobian, but is not invertible.

The conjecture remains open in two dimensions, and is easy to establish in one dimension.

The example can be stated completely explicitly: one can take

$$F(z_1, z_2, z_3) = \Big((1 + z_1 z_2)^3 z_3 + z_2^2(1 + z_1 z_2)(4 + 3z_1 z_2), \quad (1)$$
$$z_2 + 3z_1(1 + z_1 z_2)^2 z_3 + 3z_1 z_2^2(4 + 3z_1 z_2),$$
$$2z_1 - 3z_1^2 z_2 - z_1^3 z_3\Big)$$

and one can verify by a brief calculation that

$$\det DF = -2$$

and

$$F(0, 0, -1/4) = F(1, -3/2, 13/2) = F(-1, 3/2, 13/2)$$
$$= (-1/4, 0, 0).$$

While this is an extremely quick verification, the construction presented in this

A digestion of the Jacobian conjecture counterexample, 2026 · Lean PR 4474 · Isabelle AFP · arXiv 2609.17795

Correctness and novelty are separate checks

Interpreting the problem in the way suggested by Woett:
"Let $A \subseteq \mathbb{N}_0$ be a set of natural density zero. Does there exist a basis B for A such that $A \subseteq B + B$ and

I am really sorry to point this out, but in Theorem 2 of the paper of Erdos and Newman linked in this problem, a negative answer to this problem was already obtained. In particular, they showed that for most subsets A of [N] with size n, the size of smallest B with A < B + B is at least min(n log N, N^{1/2} / 2). Taking say n = N^{0.6} and gluing the sets A together for dyadic N gives the desired result.

Here is a link for the paper
https://users.renyi.hu/~p_erdos/1977-05.pdf

(The site has been updated to address this comment.)

KoishiChan — 03:39 on 25 Dec 2025 👍 3 📝 1 🤖 0

> BTW, it is really confusing why Erdos asked this again in 1980, considering his paper with Newman was published in 1977...
>
> **KoishiChan** — 03:41 on 25 Dec 2025 👍 1 📝 0 🤖 0

> This is very tragic. Back to the drawing board.
>
> **Kevin Barreto** — 03:50 on 25 Dec 2025 👍 0 📝 0 🤖 0

> > BTW I really like your work on this website. Sincerely wish you success!
> >
> > **KoishiChan** — 03:50 on 25 Dec 2025 👍 3 📝 0 🤖 0

> > > Thank you. My formal request to all members of the website is to put greater focus on literature search on the problems currently marked as open. As someone who has fallen for this twice now, it's quite gut-wrenching to find out a problem one had gotten a solution to, had already been previously solved. Of course, this is a personal error on my part, but it would help if the possibilities of such things happening was reduced overall.
> >
> > **Kevin Barreto** — 04:53 on 25 Dec 2025 👍 2 📝 0 🤖 0

JOURNAL OF NUMBER THEORY 9, 420–425 (1977)

Bases for Sets of Integers

P. ERDÖS AND D. J. NEWMAN

rom this result and 7, we obtain

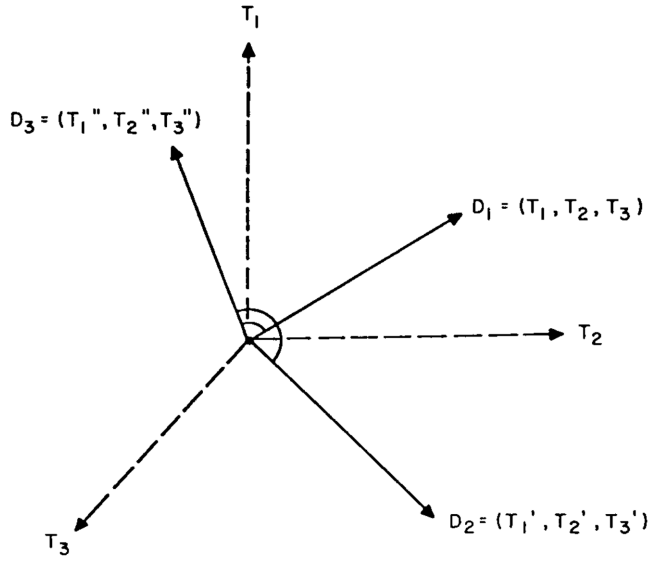
HEOREM 2. *Most sets A, of type (n, N) satisfy* $m_A > \min(n/\log N,$
$/2)$. *If furthermore, we have* $N \geqslant n^{2+\epsilon}$, $\epsilon > 0$, *then the log N may be* $\text{aced by } (1+\epsilon)/\epsilon$.

The proof is correct. The novelty claim lost within hours, when a commenter pointed to a 1977 theorem. Knowing the earlier work **changes what we can claim** as a contribution.

Erdős problem 333, forum thread · Bases for sets of integers, J. Number Theory 1977, Theorem 2

Finding related text and checking novelty are different tasks

Fig. 1. Vector representation of document space.



Weighting by specificity

This clearly suggests a weighting scheme. In normal term co-ordination matches, if a request and document have a frequent term in common, this counts for as much as a non-frequent one; so if a request and document share three common terms, the document is retrieved at the same level as another one sharing three rare terms with the request. But it seems we should treat matches on non-frequent terms as more valuable than ones on frequent terms, without disregarding the latter altogether. The natural solution is to correlate a term's matching value with its collection frequency. At this stage the division of terms into frequent and non-frequent is arbitrary and probably not optimal: the elegant and almost certainly better approach is to relate matching value more closely to relative frequency. The appropriate way of doing this is suggested by the term distribution curve for the vocabulary, which has the familiar Zipf shape. Let $f(n)=m$ such that $2^{m-1} < n <= 2^m$. Then where there are N documents in the collection, the weight of a term which occurs n times is $f(N) - f(n) + 1$. For the Cranfield collection with 200 documents, for example, this means that a term occurring ninety times has weight 2, while one occurring three times has weight 7.

Similarity finds relevant text. It does not by itself check what a paper adds, and a collection that lacks the earlier work cannot return it. Six models refilling 132,000 citation slots wrote fewer contrasting citations than the human authors and rewrote "prior work fails at X" as supportive far more often than the reverse, with or without web search (EMNLP 2026).

Vector space model, CACM 1975 · Term specificity, J. Doc. 1972 · Citing less critically, EMNLP 2026

Outline

- A. what supports confidence in an AI answer?
- B. checking claims beyond the literature
- C. why still read?
- D. reading with AI

Reading helps us build contextualized map of a field

We can follow the efficient attention paper's AI read, but may not be able to follow this quantum-chip paper.

> What background would make this explanation understandable?
>
> What would then be needed to assess its claims?

Read https://www.nature.com/articles/s41586-024-07371-7. How convinced are you by the proposed method?

Read a page

I should flag upfront what I actually got: the main text is paywalled, so what I read was the abstract, the extended data figure captions, the author contributions, and the data/code availability notes. That's more informative than it sounds — the extended data captions carry a lot of the quantitative substance — but it means I'm assessing the architecture partly through its supplementary material rather than the authors' own framing of their results.

The paper is Li et al., *Nature* 630, 70–76 (2024), from Englund's group at MIT. The proposal is a "quantum system-on-chip": diamond microchiplets containing tin-vacancy (SnV^-) color centers, transferred onto a cryogenic CMOS ASIC that supplies microwave control and electrostatic strain tuning, with spectral tuning used to give all-to-all connectivity across frequency channels.

What I find convincing. The lock-and-release transfer is the real contribution. Etching oxide locking structures into the CMOS die so diamond chiplets seat deterministically, rather than serially pick-and-placing each one, is the kind of fabrication trick that actually transfers to other labs and other material systems. The CMOS post-processing sequence in Extended Data Fig. 1 is unglamorous and well documented. Reusing the existing metal stack for both microwave delivery and capacitive strain tuning is efficient. And SnV^- is the right emitter choice — inversion symmetry buys first-order insensitivity to electric field noise, and it works at temperatures a dilution-fridge-free system can reach.

Spin-photon interfaces on a CMOS platform, Nature 2024 · ChatGPT reading · Claude reading

Four-question recipe for engineering papers

MOTIVATION why should we solve this problem

SOLUTION how did we solve (method); how do we know it's any good (eval)

OLD IDEA how did people before us attempt to solve this problem

DELTA what have we done differently

Typical exceptions:

- a phenomenon paper (no solution yet);
- a many-ancestor paper (placement is a graph, the recipe can hold one edge);
- a survey paper (a whole concept graph, which we ultimately want to build for ourselves for other fields, the recipe for every edge);

The four questions have a lineage

The Heilmeier Catechism

1. What are you trying to do? Articulate your objectives using absolutely no jargon.
2. How is it done today, and what are the limits of current practice?
3. What is new in your approach and why do you think it will be successful?
4. Who cares? If you are successful, what difference will it make?
5. What are the risks?
6. How much will it cost?
7. How long will it take?
8. What are the mid-term and final "exams" to check for success?

Who was George Heilmeier?

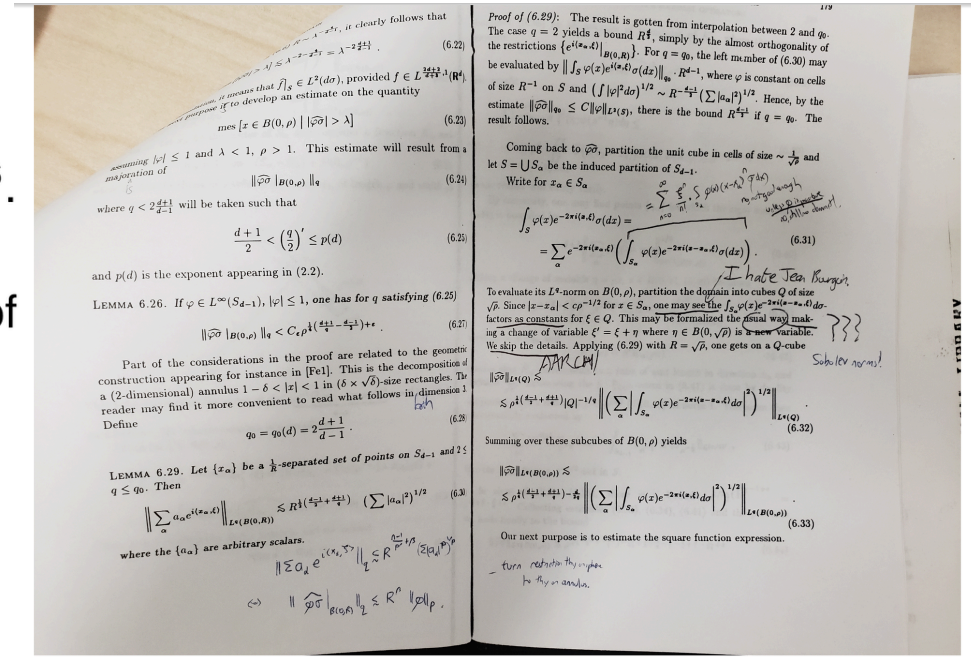
Heilmeier, who ran DARPA from 1975-1977, developed this questionnaire based on his extensive career in technology R&D. His early work at RCA is credited with laying the foundation for liquid crystal displays. After his tenure at DARPA, he served as an executive at Texas Instruments during a time when the company dominated the home computer market.

> A proposal-writing tool, repurposed for **claim auditing**.
> (this repurposing is our delta.)

The Heilmeier Catechism, DARPA

Reading earlier work also helps us choose a question

- Current AI tools have a very mixed record with proof exposition.
- On the one hand, the spelling, grammar, and formatting is close to flawless[6].
- However, the writing often dwells at length on trivialities, while passing very briefly through (or even obscuring) the most interesting and novel portions of the argument.
- AI-generated texts also often fail to note connections with prior literature, or provide high level overviews of the result.



A 1991 paper of Bourgain, annotated by my much younger self.

Which step carries the contribution, and how does it depend on earlier results?

Mathematics in the age of AI, ICM public lecture 2026, pp. 30 to 33

What we generate, we remember

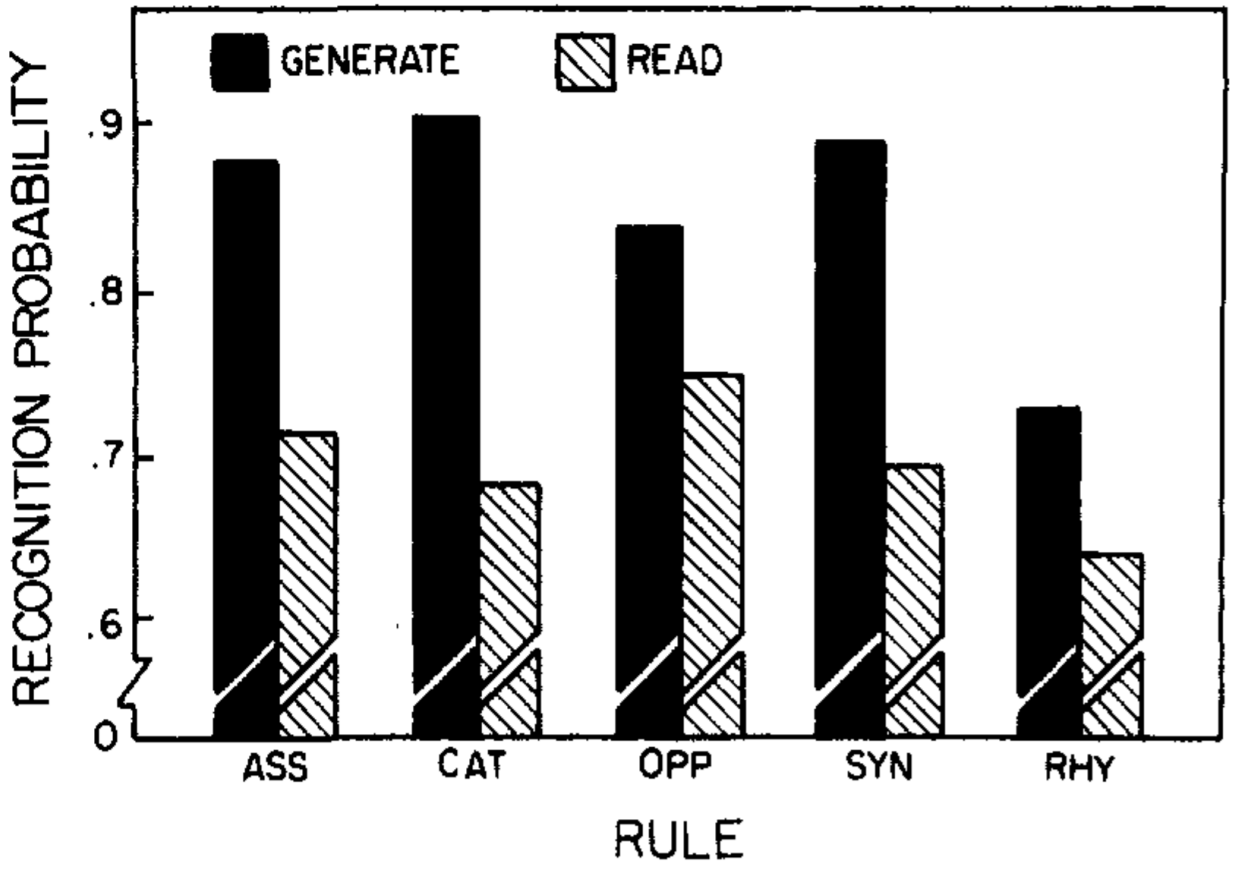


Figure 1. Mean recognition probabilities for each condition for each rule of Experiment 1. (ASS = associate; CAT = category; OPP = opposite; SYN = synonym; RHY = rhyme.)

Words we generate are remembered better than words we read (1978). Single-word recognition, not paper reading; it supplies a hypothesis for writing our own explanation before asking the model.

Distinguish the ease of producing a note from **what we can explain later**.

Why we borrow "lab checkoff' idea from 6.390 and sister courses

The generation effect, JEP:HLM 1978

Outline

- A. what supports confidence in an AI answer?
- B. checking claims beyond the literature
- C. why still read?
- D. reading with AI

What seems to work well:

1. Skim, for an **initial impression and explanation** without AI.

2. Ask AI to challenge our answer, or help with the step we cannot follow.
Check AI response against the paper, the field, the code, etc.

3. Explain the result to a partner (or your good old rubber duck) **with AI closed**.

we will practice this in lab2

Student Reading Examples (of individual papers)

1. asks for details and explanations; fact check each round and redirect AI when it drifts.

2. Write your summary of the paper and let AI read your summary and critique

3. Ask AI to quiz you on the paper you read

Why this matters

- It lets him apply matrix-style operations (like distance products)
- Missing edges can just be treated as having weight $+\infty$

One-line summary

Same formula, but now extended from **edges only → all vertex pairs**, which is the "abuse of notation."

no, i'm talking about the part where it says matrix C = {c_ij}_ij where c_ij = mink(a_ik + b_kj)

Short answer:

The "abuse of notation" is that he switches from **matrix entries** to thinking of them as **functions over points**—even though the formula looks identical.

What actually changes

- Before:

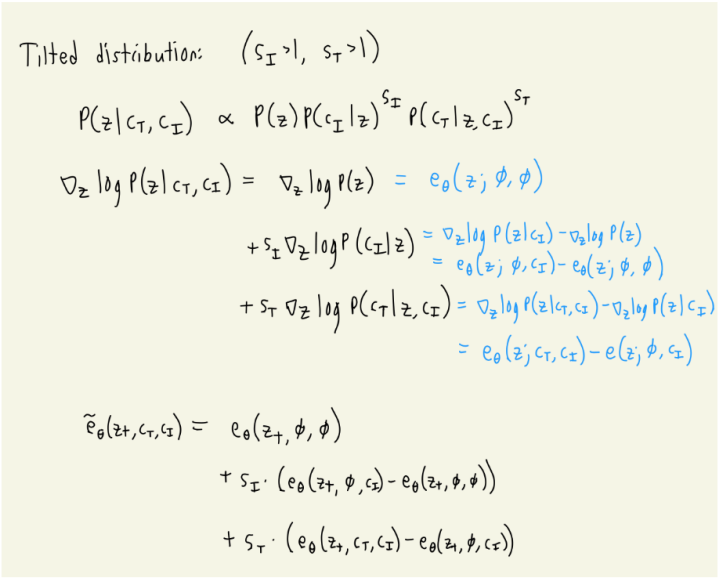
$$C = \{c_{ij}\}, \quad c_{ij} = \min_k (a_{ik} + b_{kj})$$

  → purely **matrix definition**
- After:
  - He treats rows i as **points** p_i **in** $\mathbb{R}^d$

Classifier-Free Guidance

During training, make sure to have all pairs of

(condition on text vs unconditional on text) x (condition on image vs unc



- brief explanation: recall that the diffusion model predicts a conditic

So then you can do this 2D CFG, improving adherence to both the text conditioning and the image conditioning.

(They do this derivation in Appendix B, but I think it's not super clear when they did it there)

Our generative model learns $P(z|c_I, c_T)$, the probability distribution of image latents $z = \mathcal{E}(x)$ conditioned on an input image c_I and a text instruction c_T. We arrive at our particular classifier-free guidance formulation by expressing the conditional probability as follows:

$$P(z|c_T, c_I) = \frac{P(z, c_T, c_I)}{P(c_T, c_I)} = \frac{P(c_T|c_I, z)P(c_I|z)P(z)}{P(c_T, c_I)}$$

Diffusion models estimate the score [23] of the data distribution, i.e., the derivative of the log probability. Taking the logarithm gives us the following expression:

$$\begin{aligned}\log(P(z|c_T, c_I)) = \log(P(c_T|c_I, z)) + \log(P(c_I|z)) \\ + \log(P(z)) - \log(P(c_T, c_I))\end{aligned}$$

Taking the derivative and rearranging we attain:

$$\begin{aligned}\nabla_z \log(P(z|c_T, c_I)) = \nabla_z \log(P(z)) \\ + \nabla_z \log(P(c_I|z)) \\ + \nabla_z \log(P(c_T|c_I, z))\end{aligned}$$

As we read more papers, keep connections

When the user is building a trail, he names it, inserts the name in his code book, and taps it out on his keyboard. Before him are the two items to be joined, projected onto adjacent viewing positions. At the bottom of each there are a number of blank code spaces, and a pointer is set to indicate one of these on each item. The user taps a single key, and the items are permanently joined. In each code space appears the code word. Out of view, but also in the code space, is inserted a set of dots for photocell viewing; and on each item these dots by their positions designate the index number of the other item.

Thereafter, at any time, when one of these items is in view, the other can be instantly recalled merely by tapping a button below the corresponding code space. Moreover, when numerous items have been thus joined together to form a trail, they can be reviewed in turn, rapidly or slowly, by deflecting a lever like that used for turning the pages of a book. It is exactly as though the physical items had been gathered together from widely separated sources and bound together to form a new book. It is more than this, for any item can be joined into numerous trails.

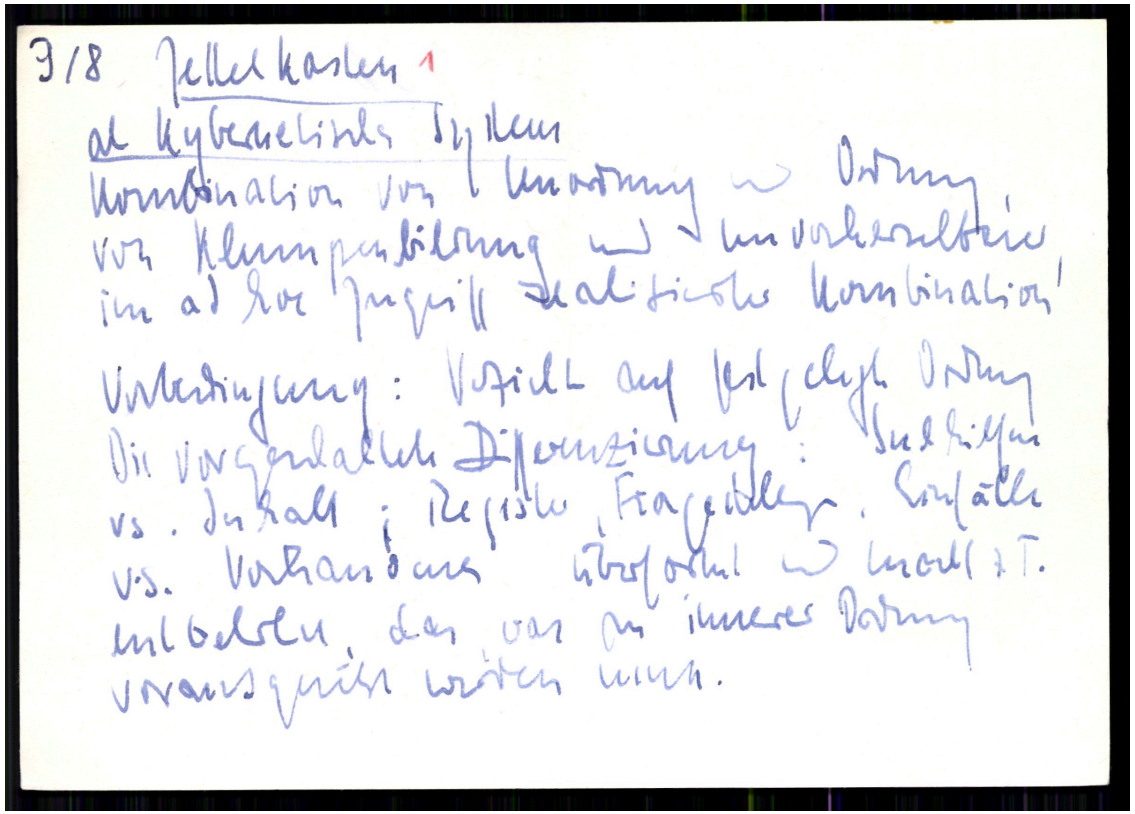
And his trails do not fade. Several years later, his talk with a friend turns to the queer ways in which a people resist innovations, even of vital interest. He has an example, in the fact that the outranged Europeans still failed to adopt the Turkish bow. In fact he has a trail on it. A touch brings up the code book. Tapping a few keys projects the head of the trail. A lever runs through it at will, stopping at interesting items, going off on side excursions. It is an interesting trail, pertinent to the discussion. So he sets a reproducer in action, photographs the whole trail out, and passes it to his friend for insertion in his own memex, there to be linked into the more general trail.

people dreamed of a research machine that stores links between documents so a reader could (re)follow a useful path as they read.

> The connection graph is still **the reader's own artifact**.
> Our lineage map is one.

As We May Think, The Atlantic, July 1945

Trails, by hand: 90,000 slips over four decades



1952 to early 1997; roughly 45,000 cross-references between them.

What would making, linking, and following these require from the reader?

Niklas Luhmann-Archiv, Zettelkasten (CC BY-NC-SA 4.0)

A wiki maintained by an LLM

Indexing and logging

Two special files help the LLM (and you) navigate the wiki as it grows. They serve different purposes:

index.md is content-oriented. It's a catalog of everything in the wiki — each page listed with a link, a one-line summary, and optionally metadata like date or source count. Organized by category (entities, concepts, sources, etc.). The LLM updates it on every ingest. When answering a query, the LLM reads the index first to find relevant pages, then drills into them. This works surprisingly well at moderate scale (~100 sources, ~hundreds of pages) and avoids the need for embedding-based RAG infrastructure.

> Incorporate sources into pages, flag contradictions, file answers back. Connections compile into pages that later questions can reuse. A workflow proposal; nothing about accuracy or learning is measured. When a tool uses saved notes as memory, check what it actually receives: in an audit of donated ChatGPT histories, 96% of memory entries were written by the system rather than asked for (WWW 2026).

LLM Wiki gist, 2026-04-04 · The algorithmic self-portrait, WWW 2026

A wiki maintained by an LLM

LLM Wiki

A pattern for building personal knowledge bases using LLMs.

This is an idea file, it is designed to be copy pasted to your own LLM Agent (e.g. OpenAI Codex, Claude Code, OpenCode / Pi, or etc.). Its goal is to communicate the high level idea, but your agent will build out the specifics in collaboration with you.

The core idea

Most people's experience with LLMs and documents looks like RAG: you upload a collection of files, the LLM retrieves relevant chunks at query time, and generates an answer. This works, but the LLM is rediscovering knowledge from scratch on every question. There's no accumulation. Ask a subtle question that requires synthesizing five documents, and the LLM has to find and piece together the relevant fragments every time. Nothing is built up. NotebookLM, ChatGPT file uploads, and most RAG systems work this way.

The idea here is different. Instead of just retrieving from raw documents at query time, the LLM **incrementally builds and maintains a persistent wiki** — a structured, interlinked collection of markdown files that sits between you and the raw sources. When you add a new source, the LLM doesn't just index it for later retrieval. It reads it, extracts the key information, and integrates it into the existing wiki — updating entity pages, revising topic summaries, noting where new data contradicts old claims, strengthening or challenging the evolving synthesis. The knowledge is compiled once and then *kept current*, not re-derived on every query.

This is the key difference: **the wiki is a persistent, compounding artifact.** The cross-references are already there. The contradictions have already been flagged. The synthesis already reflects everything you've read. The wiki keeps getting richer with every source you add and every question you ask.

The gist says the LLM owns the wiki layer and the human curates sources. It also cites the memex, "the part he couldn't solve was who does the maintenance."

LLM Wiki gist, 2026-04-04 · The algorithmic self-portrait, WWW 2026